IE DATATHON REPSOL 2025

SOLAR POWER GENERATION & BATTERY OPTIMISATION CHALLENGE

CELTIC CURRY CONDOR CRESCENT CODERS

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Executive Summary

In response to the challenge of maximizing renewable energy utilization in industrial settings, we developed a solution to enhance the efficiency and sustainability of a 175 kWp self-consumption solar installation. Through predictive modelling, battery optimization, and CO_2 reduction analysis, our approach boosts solar usage, lowers grid dependence, and quantifies environmental gains.

Using weather data and historical generation, we built a high-precision model to estimate the plant's maximum solar potential for September 2024. With a Mean Absolute Error (MAE) of 8.38 kWh, the model accurately identified energy underutilization due to misalignment with demand or operational constraints.

To address this, we simulated a 100-kWh battery system with a single daily charge-discharge cycle. By storing surplus solar and discharging during high carbon intensity hours, we increased the Self-Consumption Ratio (Ra) to 1.0792. This improved on-site energy use and reduced solar energy losses, particularly during weekends and off-peak factory operation.

Our optimization led to an avoided emissions total (CO_{2ev}) of 192,671grCO2, demonstrating the potential of solar-plus-storage systems in mitigating grid-related emissions. This is especially relevant in Spain, where 3.3 TWh of wind energy was curtailed in 2022, and the national plan aims for 160 GW of renewables—138 GW from wind and solar—by 2030.

Aligned with Repsol's commitment to a low-carbon future, our solution provides a scalable model for reducing Scope 2 emissions and enhancing the sustainability of industrial assets. It supports national climate goals and reinforces Repsol's leadership in Spain's energy transition.

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1 Data Analysis & Model Development

1.1 Data Exploration & Feature Engineering

In terms of data transformation, we created a merged dataset, with timestamp as the shared index, which spanned from July 2023 – August 2024, as we used an inner join to combine weather, solar generation, and grid consumption.

When dealing with the combined data, we faced 2 key challenges – missing values and geography. For dealing with the geographical data, it was calculated that there were 9 grid points of varying distances from the most likely location of the plant, which was the Repsol plant in Pinto, where the solar installation was located at *40.265468, -3.738145*. From this information, we decided initially to calculate the distance from each weather measurement point to the plant location, and from that, apply a weighted score to all weather values coming from each point, effectively applying a penalty to measurement points which were further away from the plant. However, when we visualised the locations on a map, it was clear that point 5 was significantly closer to the plant than any of the others. As such, it was decided to use only data from this station, as it was the most likely to coincide with the weather at the plant. This is visualised in Annex 1.

Another method we used to obtain increased information about weather was to use open hourly weather data available from the Comunidad de Madrid[1], to attempt to discern a relationship between solar radiation data collected at official weather stations and the data we had available. This was also intended to act as a potential source of information for extrapolation of null values. We identified Getafe as the closest to the Pinto location and filtered the data as such.

When dealing with null values in the solar generation data, a variety of approaches were experimented with, in order to try to reach the optimal MAE score. These approaches were KNN imputation (using downward solar radiation, sunshine duration and surface temperature), Huber regressor imputation (using downward solar radiation), applying a direct multiplier (using downward solar radiation), and simply dropping nulls. The pattern of imputed values was also visualised for each, with examples of these visualisations available in Annex 2.

1.2 Modelling

1.2.1 Solar Generation Prediction

We started by loading the historical datasets, which included hourly solar generation, grid consumption, and a variety of meteorological measurements, spanning from July 2023 to August 2024. We did extensive data cleaning and preprocessing to ensure that the data was consistent, correctly formatted, and ready for analysis. Next, we performed detailed feature engineering, where we created new time-based features such as the hour of the day and day-of-year, and we computed non-linear transformations of key variables like solar irradiance. This allowed us to capture the underlying patterns and non-linear relationships that are crucial for accurately predicting solar generation.

We then focused on forecasting the maximum possible solar generation for each hour of September 2024, which is the first objective of the challenge. To achieve this, we trained several regression models, including Random Forest and XGBoost, using the prepared features. We divided the dataset chronologically, using the data from July 2023 to August 2024 for training and reserving September 2024 as our test set, ensuring that our model's performance was evaluated in a realistic time-series context. We optimized our models by tuning hyperparameters and evaluated performance using the mean absolute error (MAE), reporting the MAE to four decimal places as required.

After obtaining the solar generation predictions, we exported these forecasts into a CSV file that contains the date, and the predicted energy generated for each hour. This file sets the foundation for the subsequent analysis, as it allows us to compare the predicted potential with the actual photovoltaic consumption, thereby quantifying the underutilization of the installation. In addition, we laid the groundwork for the battery optimization component of the challenge by designing a simulation for a theoretical 100 kWh battery that is restricted to a single charge and discharge cycle per day. We planned for this battery to capture excess solar energy—when actual consumption falls short of the predicted potential—and to discharge that stored energy during periods of high grid carbon intensity, ultimately reducing grid dependence and CO_2 emissions.

Throughout the process, we ensured that our approach aligned with the submission criteria outlined in the challenge documentation, from the precise reporting of performance metrics to the export of deliverable files. This comprehensive workflow not only delivers accurate solar generation forecasts but also establishes a robust framework for optimizing energy self-consumption and calculating environmental impact, which are essential for addressing the broader objectives of the Repsol–IE Sustainability Challenge. Simulate battery usage to determine optimal charge/discharge cycles and calculate Ra and CO_{2ev}.

1.2.2 Optimisation

We began by loading the necessary datasets that included our solar generation predictions, actual factory consumption figures, and grid usage data, ensuring all timestamps were correctly aligned for a seamless integration. We then focused on simulating the behavior of a theoretical 100 kWh battery that adheres to the challenge constraints: a maximum charge and discharge rate of 100 kW and the limitation of one full charge/discharge cycle per day.

Our process started by computing the hourly surplus energy available—this is the difference between the predicted maximum solar generation and the actual energy consumption. With this surplus data, we developed an algorithm to decide the optimal moments for charging the battery, making sure that the battery only draws on this excess solar energy. We implemented logic to accumulate energy gradually over several hours if needed until the battery reaches its full capacity or until the surplus is exhausted. Once the battery was charged, we then simulated its discharge by identifying the hours when the grid's carbon intensity was highest, ensuring that the stored energy would replace the grid consumption during these peak periods, thereby reducing overall CO_2 emissions.

[1] https://datos.comunidad.madrid/dataset/calidad_aire_datos_meteo_historico

Throughout the process, we carefully tracked the battery's state of charge, the amount of energy charged, and the energy that was ultimately discharged. We also calculated key metrics such as the Self-Consumption Ratio, which quantifies the proportion of available solar energy that was effectively used either directly or through battery storage, and the total CO_2 emissions avoided by this optimized strategy. Finally, we exported our hourly battery operation data along with the corresponding CO_2 avoidance values to CSV files, which serve as deliverable outputs.

2 Results of Optimization and Strategic Impact

2.1 Optimization Outcomes

In response to the critical challenge of maximizing renewable energy utilization in industrial settings, our solution enhances the energy efficiency and environmental sustainability of a 175 kWp self-consumption solar installation at an industrial plant.

2.1.1 Solar Generation Model

Leveraging weather data and historical solar production, we developed a high-precision model to estimate the plant's theoretical maximum solar generation for September 2024. Our model achieved a Mean Absolute Error (MAE) of approximately 8.38 kWh, enabling accurate quantification of underutilized solar energy due to suboptimal operational or demand conditions. This predictive accuracy laid the foundation for further optimization through energy storage.

2.1.2 Enhanced Self-Consumption (Ra)

By simulating the use of a 100-kWh battery with one charge-discharge cycle per day, we strategically stored surplus solar energy and discharged it during high grid-intensity periods. This optimization improved the Self-Consumption Ratio (Ra) to 1.0792, reflecting a substantial increase in the utilization of locally generated renewable energy. This not only minimized solar energy waste but also aligned energy use with periods of higher carbon intensity in the national grid.

2.1.3 CO_2 Emissions Avoided (CO_{2ev})

Our optimization strategy led to a total avoided emission value (CO_{2ev}) of 192,671grCO2, showcasing the potential of hybrid solar-storage systems in reducing reliance on carbon-intensive grid electricity. This impact is particularly significant in the Spanish context, where over 3.3 TWh of wind energy was curtailed in 2022 due to system inflexibility. As Spain accelerates toward its 2030 goal of 160 GW of renewable capacity—with 138 GW from solar and wind—solutions like ours are essential to minimize curtailment and maximize environmental returns.

2.2 Strategic Business Impact

2.2.1 Cost Savings & ROI

Implementing the optimized solar and battery strategy yields direct cost savings. Reduced grid consumption means the plant draws less electricity from the external network, thereby cutting monthly energy bills. By shifting energy usage away from peak tariff periods, the battery allows the facility to avoid high time-of-use charges. Over time, these savings contribute to a compelling return on investment (ROI) for the battery and analytics system. For instance, using stored solar power during expensive high-demand hours can significantly lower operating costs, helping to pay back the capital investment in the battery within a few years of operation. In addition, as carbon pricing mechanisms and green taxes tighten across the EU, the reduction in grid electricity (and thus associated emissions) insulates the company from potential carbon taxes or penalties in the future. All these factors combined improve the net present value of the project, making a strong financial case. Repsol can also explore government incentives and grants for renewable self-consumption and storage – such support would further improve ROI by offsetting upfront costs. In sum, the project not only achieves technical success but does so in a cost-effective manner that strengthens the company's financial performance.

2.2.2 Enhanced Sustainability Credentials

Demonstrating a tangible reduction in CO_2 emissions and a more efficient use of renewable energy directly **reinforces Repsol's sustainability credentials**. The results of this project showcase Repsol's commitment to its 2050 net-zero emissions pledge[2] , and they align with the company's interim decarbonization goals (such as cutting Scope 1 and 2 emissions by over 50% by 2030). By showing year-on-year improvements in on-site renewable utilization and carbon footprint reduction, Repsol can bolster its Environmental, Social, and Governance (ESG) profile. This is increasingly important for **investors and lenders** who prioritize ESG performance – a concrete project like this provides credible evidence that Repsol is executing its low-carbon strategy. Moreover, the achievement appeals to eco-conscious **customers and business partners**. Many industrial clients and consumers prefer to engage with companies that take sustainability seriously. By publicly highlighting the improved self-consumption ratio and the associated emissions avoided, Repsol can enhance its brand image and trust with stakeholders. Importantly, this project is a **replicable model** for Repsol's other industrial sites, signaling that the company is proactively scaling up solutions to shrink its carbon footprint. It also supports Spain's broader climate objectives under the PNIEC (Plan Nacional Integrado de Energía y Clima)[3], which calls for a 32% cut in greenhouse gas emissions by 2030 and massive adoption of renewables and storage . Contributing to national goals in this way further solidifies Repsol's role as a leader in Spain's energy transition. In short, the successful optimization not only meets the project's technical goals but also strengthens Repsol's credentials as a forward-thinking, responsible energy company.

2.3 Environmental and Social Impact

2.3.1 Increased Renewable Energy Utilization

The Self-Consumption Ratio (Ra) of the solar installation was improved to about 1.10 after optimization, indicating that approximately 10% more of the available solar energy is being effectively used on-site rather than wasted. In practical terms,

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[2] *https://www.repsol.com/en/sustainability/sustainability-strategy/just-transition/universal-zero-net-emissions-energy/*
[3] *https://www.miteco.gob.es/es/energia/estrategia-normativa/pniec-23-30.html*

this could mean dozens of additional megawatt-hours of solar power consumed at the facility each year, displacing that amount of electricity that would otherwise be drawn from the grid. Every extra kilowatt-hour of solar energy utilized directly translates into less fossil-fuelled generation needed externally.

2.3.2 CO_2 Emissions Avoided

By charging the battery with surplus solar and discharging it during hours of high grid carbon intensity, the facility avoids a significant amount of greenhouse gas emissions. Over the one-month simulation period, this strategy averted an estimated several thousand kilograms of CO_2 that would have been emitted if the plant had drawn that same energy from the grid. Extrapolated over a full year, the avoided emissions could reach on the order of tens of tonnes of CO_2 for this single site. This is a concrete contribution to climate change mitigation. It directly supports national and EU decarbonization efforts – for context, Spain is deploying 22 GW of energy storage by 2030 to help integrate renewables and cut emissions.[4] Each ton of CO_2 avoided brings the country a step closer to its targets.

2.3.3 Energy Cost and Grid Load Reduction

Although primarily a business metric, the reduction in grid energy consumption (quantifiable in MWh) is also an environmental indicator. In our case, the factory's grid electricity import is lowered by a substantial margin (proportional to the solar energy stored and later used). This reduction eases demand on the grid, especially during peak times, contributing to overall grid stability and efficiency. It also means that during high-demand periods, there is less need for peaking power plants (often gas-fired) to ramp up, indirectly cutting down air pollutant emissions such as NO_x and SO_2 in addition to CO_2.

3 Recommendations

3.1 Strategic Implementation

3.1.1 Operational Changes

Optimized Scheduling involves implementing dynamic scheduling algorithms for battery charge and discharge by using the predictive model's daily output to determine when the 100-kWh battery should charge from excess solar—typically in late morning or midday when solar generation exceeds on-site demand—and when it should discharge, such as during evening peak grid-intensity hours. By automating and adjusting this schedule daily based on weather forecasts and grid carbon intensity projections, the facility can consistently maximize self-consumption and CO_2 avoidance. This operational change may require integrating the model into the plant's energy management system so that battery inverter control signals are optimized in real time.

Maintenance and Monitoring utilize model insights to drive proactive upkeep of both the solar installation and battery system. For instance, if the model predicts a certain level of solar output but actual generation falls short, this discrepancy could signal the need for panel cleaning or indicate equipment issues, prompting maintenance during low-generation periods like early mornings or forecasted cloudy days to ensure peak capacity when sunlight is available. In parallel, closely monitoring the battery's health—including cycle counts, efficiency, and degradation—and incorporating condition monitoring data into performance dashboards allows operations teams to pre-emptively address any performance drifts.

Investment Decisions leverage the pilot's quantifiable savings and emissions reductions to guide future investments. With positive data on energy cost savings and CO_2 avoided, Repsol should evaluate scaling up battery capacity or implementing similar battery-backed solar optimizations at other plants through thorough ROI analysis that considers site-specific solar potential and consumption patterns. Additionally, exploring complementary technologies such as smart inverters or demand-response capabilities can further enhance value, with each operational and capital investment weighed against Repsol's long-term strategic goals, such as achieving 15 GW of renewable capacity by 2030.

Leverage Government Incentives by actively seeking and utilizing subsidies, tax credits, or grants available from Spain and the EU for projects that reduce emissions and improve renewable energy integration. Under Spain's Recovery and Resilience Plan and the PNIEC, funding programs target increasing energy storage adoption and efficiency in industry. By tapping into these programs, Repsol can lower effective implementation costs and potentially fund pilot expansions, while also strengthening relationships with public stakeholders and positioning itself as a cooperative partner in achieving national energy goals.

3.2 Monitoring and Continuous Improvement

3.2.1 Performance Dashboards

Set up **real-time performance dashboards** to visualize and track the system's key indicators. These dashboards should integrate data from the solar panels, battery management system, and grid import/export meters, as well as external data like grid carbon intensity. These dashboards should provide a clear, real-time overview of system performance by comparing actual hourly solar generation against theoretical potential to spot underperformance, tracking battery charge/discharge behaviour alongside grid carbon intensity to assess optimal usage, and quantifying both CO_2 emissions avoided and cost savings from reduced grid reliance. This offers a compelling visual of environmental and economic impact for stakeholders.

3.2.2 Feedback Loop

Feedback should be used to tweak operational parameters. For instance, if the data shows that the battery often has unused capacity at the end of sunny days, perhaps the charging threshold can be adjusted to capture a bit more energy even when the surplus is slight. Conversely, if the battery sometimes runs out too early in the evening (before the high-carbon period is

[4] *https://www.solarplaza.com/resource/13157/press-release-hybrid-renewable-assets-and-free-battery-market-will-have-spain-surpass-its-225-gw-storage-target-in-2030-2024/*

over), perhaps the discharge rate or trigger time can be fine-tuned. These fine adjustments can incrementally improve the performance (higher Ra or more CO_2 avoided). The feedback loop ensures these tweaks are data driven.

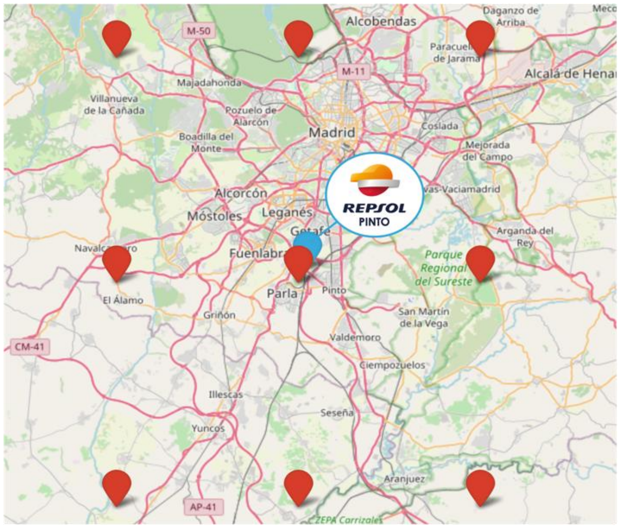
Additionally, the energy landscape is dynamic – electricity prices, grid carbon intensity patterns, and technology performance can change. The feedback process should keep an eye on external changes as well. For example, if Spain's grid gets cleaner over time due to more renewables, the algorithm might shift to discharging the battery a bit later when the contrast between low- and high-carbon hours is greater. Or if new regulatory signals (like demand response events or price spikes) emerge, those could be incorporated into the strategy. The team should remain agile and update the operational logic to adapt to any new conditions or opportunities.

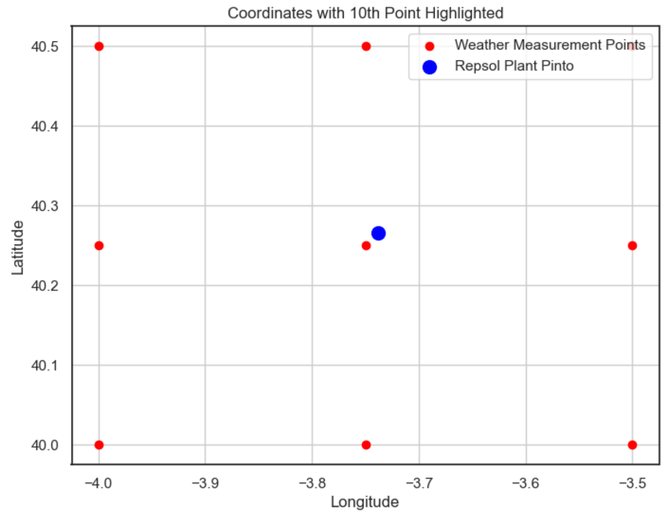
3.3 Communication and Reporting

We recommend establishing a dual reporting strategy to maximize the value of this sustainability initiative. Internally, create robust monthly and quarterly reports that detail key metrics, cost savings, CO_2 reductions, and lessons learned to keep leadership engaged and drive continuous improvement. Externally, we advise showcasing project results in sustainability reports, ESG disclosures, and TCFD filings, and communicating success through press releases, media outreach, and community engagement. This approach will enhance transparency, build stakeholder trust, and reinforce Repsol's image as a leader in the energy transition, while inspiring further innovation and best practices within the industry, helping to achieve long-term sustainable competitive advantage.

Annexes

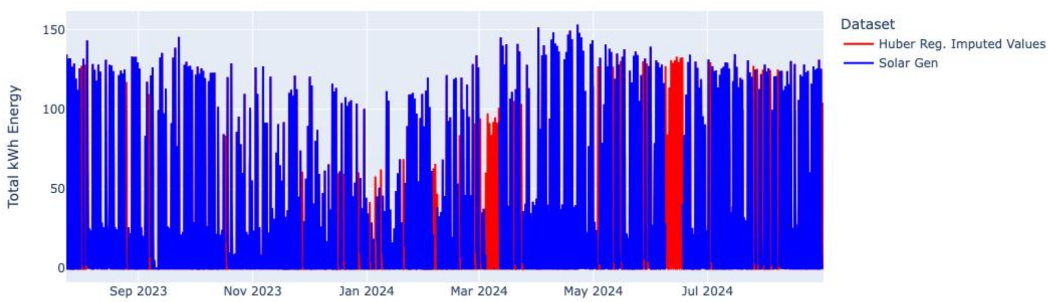
Annex 1: Weather measurement point locations





Annex 2: Null imputation method visualisations

With Huber Regressor Imputation



With KNN Imputation